Task 5: Predictive Statistical Modelling

All models fit on a 75/25 train/test split (train n=20,275, test n=6,759), stratified naturally by the dataset's own 43.6% anemia prevalence (train 43.64%, test 43.56% - close enough to confirm the split is representative).

5.1 A Critical Design Decision: What Can We Predict Anemia From?

Before fitting any model, one methodological issue had to be resolved. In this dataset, anemia (Yes/No) is **deterministically derived** from HB compared against who_hb_threshold. If HB were included as a predictor, the model would trivially achieve near-perfect accuracy - not because it has learned anything, but because it is circularly re-deriving the label from the value that defines it. This would be a serious methodological flaw, not a genuine predictive modelling exercise.

We therefore predict anemia from variables that are **not used to construct the label**:

- **Demographic:** age_at_test, sex
- **Other red-cell indices:** MCH, MCHC, MCV, PCV, RBC - correlated with HB but not definitional of the anemia outcome

This simulates a genuine, defensible screening scenario: could a clinician or triage system flag likely anemia risk using demographic information and partial FBC results, without a direct HB reading? This is more realistic than it may first appear - PCV/RBC/MCV are sometimes available from point-of-care devices or partial panels before a full HB assay is confirmed, and demographic risk stratification alone is often the first triage step in resource-constrained settings.

institution is deliberately **excluded** as a predictor, consistent with the justification developed in Task 1 and confirmed statistically in Task 4 (Test E): the institution effect is real but weak (Cramer's V = 0.093), concentrated in one facility (Institution_C), and confounded with severe sample-size imbalance - including it would harm generalizability without meaningfully improving genuine predictive signal.

5.2 Model Selection and Justification

#	Model	Why included
1	Logistic regression - demographic only	Baseline: how much can age/sex alone tell us?
2	Logistic regression - demographic + labs	Adds the other FBC indices as predictors
3	Logistic regression - demographic + labs + age×sex interaction	Tests whether Task 4's confirmed interaction effect improves prediction
4	Ridge regression (glmnet, α=0)	Addresses multicollinearity via L2 shrinkage, keeps all predictors
5	LASSO regression (glmnet, α=1)	L1 shrinkage; performs automatic variable selection
6	Elastic Net (glmnet, α=0.5)	Compromise between Ridge and LASSO

Logistic regression is the correct base model because the outcome (anemia) is binary - linear regression would be inappropriate (unbounded predictions, non-constant error variance for a 0/1 outcome). Logistic regression is itself a **Generalized Linear Model**: binomial family, logit link function, satisfying the GLM requirement in this task directly while also being the most interpretable choice for a clinical/consultancy audience (coefficients convert directly to odds ratios).

5.3 Why Regularization Was Necessary, Not Just a Checklist Item

Task 3's correlation heatmap already flagged strong relationships among the red-cell indices (HB-PCV r=0.96, MCH-MCV r=0.88). Fitting Model 2 confirms this is a serious practical problem via **Variance Inflation Factors (VIF)**:

Predictor	VIF
age_at_test	1.89
sex	1.40
MCH	**387.8**
MCHC	106.5
MCV	**461.0**
PCV	40.5
RBC	59.3

Conventionally, VIF > 10 signals problematic multicollinearity; VIF values in the hundreds (MCH, MCV) indicate the ordinary logistic regression's individual coefficient estimates for these variables are highly unstable - small changes in the data could swing them substantially, even though the model's overall predictions remain fine. **This is not a hypothetical concern for this dataset - it is a measured, confirmed problem**, which is exactly the scenario Ridge/LASSO/Elastic Net are designed to address. This elevates the regularized models from "included because the brief lists them" to "included because the data specifically requires them."

5.4 Results - Standard Logistic Regression Models

Model 1: Demographic only

Predictor	Odds Ratio	95% CI	p-value
age_at_test	1.023	[1.021, 1.024]	< 0.001
sex (Male vs Female)	0.334	[0.314, 0.355]	< 0.001

Interpretation: Each additional year of age increases anemia odds by ~2.3%; being male reduces anemia odds by ~67% relative to female, holding age constant. This mirrors Task 4's descriptive/inferential findings but offers weak standalone predictive power.

Test-set performance: Accuracy 64.2%, Sensitivity 50.5%, Specificity 74.8%, AUC ≈ 0.68 - barely better than chance-plus-prevalence. Demographic information alone is **not sufficient** for reliable screening.

Model 2: Demographic + Labs

Adding MCH, MCHC, MCV, PCV, RBC dramatically improves fit (Residual deviance drops from 25,330 to 6,738; AIC from 25,336 to 6,754).

Test-set performance: Accuracy 92.5%, Sensitivity 91.6%, Specificity 93.1%, Precision 91.1%, AUC ≈ 0.988 - a large, genuine improvement, confirming that even without HB itself, the other red-cell indices carry strong anemia-relevant signal (as expected, since they are physiologically linked to HB).

Model 3: Demographic + Labs + Age×Sex Interaction

Justified directly by Task 4's Test G, which found a statistically significant age×sex interaction effect on HB ($\eta^2 = 0.066$). We test whether this translates into improved predictive performance, not just descriptive insight.

Likelihood ratio test (Model 2 vs. Model 3): Δdeviance = 998.87 on 1 df, **$p < 2.2 \times 10^{-16}$** - the interaction term is highly statistically justified, not just marginally significant.

AIC comparison: Model 2 = 6,754.0 → Model 3 = 5,757.2 (a substantial improvement; a drop of this magnitude is decisive, not a matter of interpretation).

Test-set performance: Accuracy **94.4%**, Sensitivity 93.7%, Specificity 94.9%, Precision 93.5%, AUC ≈ 0.986 - the best-performing model in this study.

Interpretation: The significant age_at_test:sexMale interaction term (estimate = 0.101, $p < 0.001$) confirms that age's effect on anemia risk is meaningfully different for men and women - consistent with, and now doubly confirmed by, both the inferential (Task 4) and predictive (Task 5) analysis. This is a genuinely convergent, well-triangulated finding across two different statistical approaches.

5.5 Results - Regularized Models

Using the same predictor set (excluding HB and institution), 10-fold cross-validation selected the optimal penalty (λ) for each method.

Model	Accuracy	Sensitivity	Specificity	Precision	AUC (approx.)
Ridge (α=0)	92.0%	89.4%	94.0%	92.0%	0.979
LASSO (α=1)	92.4%	91.4%	93.2%	91.2%	0.983
Elastic Net (α=0.5)	92.4%	91.5%	93.2%	91.2%	0.983

Key finding - LASSO performs automatic feature selection: at the cross-validated optimal λ, LASSO shrank the coefficients for **MCH and RBC to exactly zero**, effectively removing them from the model. This is not arbitrary - MCH is the variable most collinear with MCV (r=0.88, VIF=387.8) and RBC is highly collinear with PCV (r=0.82, VIF=59.3). LASSO independently identified and removed the same redundant variables that the VIF diagnostic flagged, which is a reassuring cross-check between two different diagnostic approaches.

Ridge, as expected, shrank all coefficients toward zero but retained every predictor (L2 penalty does not force exact zeros) - appropriate when the analyst wants to keep all clinically meaningful variables in the model despite collinearity, at the cost of individual coefficient interpretability.

5.6 Model Comparison and Assumption Discussion

Model	Accuracy	AUC	Interpretability	Handles multicollinearity?
1. Logistic (demo only)	64.2%	0.68	High	N/A (no collinear predictors)
2. Logistic (demo+labs)	92.5%	0.988	High, but unstable coefficients	No - VIF up to 461
3. Logistic (demo+labs+interaction)	**94.4%**	0.986	High	No - same issue as Model 2
4. Ridge	92.0%	0.979	Moderate (shrunk, all retained)	Yes
5. LASSO	92.4%	0.983	Moderate (some removed)	Yes
6. Elastic Net	92.4%	0.983	Moderate	Yes

Logistic regression assumptions relevant here:

(i) independence of observations - reasonably satisfied, as each row is a distinct patient test event, though repeated patients across multiple encounters over the 2014–2023 period are a possible mild violation worth flagging as a limitation;

(ii) linearity of the logit in continuous predictors - not formally tested here but reasonable given the smooth, monotonic-looking demographic effects observed;

(iii) absence of severe multicollinearity - **explicitly violated** in Models 2 and 3 for the raw coefficient estimates (though this does not necessarily harm predictive accuracy, only interpretability and coefficient stability).

Model 3 has the best raw accuracy, but its individual lab-index coefficients should not be over-interpreted given the VIF problem (e.g., the large, somewhat counter-intuitive positive RBC coefficient and negative MCH coefficient are partly artefacts of collinearity, not necessarily each variable's true independent effect). LASSO and

Elastic Net trade a small amount of accuracy for materially more trustworthy, stable coefficients - an important trade-off to communicate to the client rather than simply reporting "Model 3 wins."

5.7 Strengths and Limitations

Strengths:

- Excellent discrimination (AUC 0.98+) across all lab-inclusive models - strong evidence that FBC-based screening is viable even without a direct HB reading.
- The age×sex interaction, motivated by prior inferential findings (Task 4), delivers a measurable, statistically justified performance improvement - a coherent analytical narrative rather than an arbitrary model choice.
- LASSO's automatic feature selection independently corroborates the VIF-based multicollinearity diagnosis, strengthening confidence in both findings.

Limitations:

- **Multicollinearity** among red-cell indices limits how confidently individual lab-index coefficients can be interpreted causally in the unregularized models.
- The model is trained on cases with a **complete red-cell panel**; in practice, if PCV, MCV, MCH etc. are not routinely tested together, the model's real-world applicability would be more limited than these results suggest.
- **Institutional imbalance** (65% of data from Institution_B) means the model's learned patterns may reflect that institution's patient population more than the others', despite excluding institution as a predictor.
- Excluding HB is methodologically necessary here but means this cannot fully replace direct HB-based diagnosis - it is best framed as a **risk-flagging / triage support tool**, not a diagnostic replacement.
- The reported AUC uses an approximate (resampled Mann-Whitney) estimator for computational simplicity rather than exact trapezoidal ROC integration; values are consistent across repeated runs and models but should be read as indicative rather than to the fourth decimal place.

5.8 Recommendation for Organizational Use

We recommend **Model 3 (logistic regression with age × sex interaction) as the primary model for interpretability and clinical communication** - its coefficients translate directly into odds ratios that clinicians and administrators can reason about, and its interaction term is independently justified by Task 4's formal inferential test, giving the client a coherent, defensible analytical story rather than a black-box result.

However, for any **operational deployment** where the same model will be retrained periodically on new data, we recommend **Elastic Net** as the production choice: it achieves accuracy within 2 percentage points of Model 3, automatically manages the confirmed multicollinearity issue, and produces materially more stable coefficients across data refreshes.

In both cases, we recommend the client treat this as a **screening/triage support tool that flags elevated risk for follow-up**, not a diagnostic replacement for direct hemoglobin testing.